Graph Naming Standard: Insert for Semantic Governance Standards

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What changes and why

The Semantic Asset Metadata Standard already says a graph IRI equals its asset URI, that each archived release loads into a version IRI, and that the named graph and repository live on a catalog record. This insert keeps all of that and adds what the alternate-label and customer-terminology design needs: two new asset types (lexicon and mapping), two new tiers (enterprise and customer), an opaque tenant ID, reserved path segments for drafts and change records, and environment-neutral repository IDs. Because names never carry a tenant name, an environment, a version or a repository, graphs can move between repositories and servers as the topology grows without any name changing.

The wording below uses MUST and SHOULD so it can be pasted into the Standards as normative text. The example names are illustrative.

Asset names

An asset name is lowercase ASCII words joined by hyphens, in this order:

```
<type>-<tier>-<name>
<type>-customer-<tenant-id>-<name>
```

Part	Allowed values	Meaning
type	vocabulary, taxonomy, ontology, lexicon, mapping	The five kinds of asset. There are no modules. A lexicon holds labels only. A mapping holds match relations between assets.
tier	core, domain, enterprise, customer	Core is the Domain Reference Model foundation. Domain is a business domain, including operational domains. Enterprise is Insperity-wide house terms. Customer is one tenant.

Part	Allowed values	Meaning
tenant-id	`t` followed by 8 lowercase hex characters, customer tier only	Opaque, issued at provisioning, immutable and never reused. It is never the customer's name.
name	Hyphenated words; for the domain tier the first word is the domain	What the asset is about. Proposed limit: 40 characters.

Not every type exists at every tier.

Tier	Allowed types	Holds
core	vocabulary, taxonomy, ontology, mapping	Preferred labels, definitions, relations and a small vetted set of universal alternate labels
domain	vocabulary, taxonomy, ontology, lexicon, mapping	The same for one domain, plus labels common to that domain but not universal
enterprise	vocabulary, taxonomy, lexicon, mapping	House terms valid for every tenant
customer	vocabulary, taxonomy, lexicon, mapping	One tenant's labels and local concepts; never an ontology

Graph classes and IRIs

Every graph IRI derives from the asset URI. The asset URI is version-free, follows the Guide's URI pattern, and ends in the asset name.

Graph class	IRI	Holds	Lifetime
Current release	The asset URI	The released triples of the asset	Replaced by each release
Version	Asset URI + `/` + version (SemVer, starting with a digit)	One archived release	Permanent

Graph class	IRI	Holds	Lifetime
Draft	Asset URI + `/draft/` + change ID	The working copy of a proposed change	Deleted after release or rejection
Simulation	Asset URI + `/sim/` + run ID	The workspace of one what-if run, read from drafts and releases	Deleted when the run's record is closed
Change records	Asset URI + `/changes`	Append-only decisions and releases, as triples	Permanent
Catalog	Base URL + `/catalog`	Catalog records: asset, named graph, repository, tier and tenant	Permanent

The segments `draft`, `sim`, `changes` and `catalog` are reserved. A version segment always starts with a digit, so reserved words and versions cannot collide. A simulation never writes to a current-release or version graph.

Repository IDs

A repository ID is environment-neutral, as the Metadata Standard already requires. The form is `skl-` plus a purpose: `skl-core-domain`, `skl-lexicons`, `skl-extensions`, and `skl-tenant-` plus the tenant ID for a dedicated repository. The environment (dev, test, stage or prod) belongs to the server address and never to a repository or graph name. Moving an asset between repositories changes only its catalog record, which is what lets the topology grow without renaming anything.

Tenant IDs

The catalog maps a tenant ID to its organization, and only the tenant provisioner and auditor personas can read that mapping. The organization's name never appears in an IRI, a log line or a change record.

Rules

1. An asset name MUST use only lowercase ASCII letters, digits and hyphens, and MUST follow `<type>-<tier>-<name>` or the customer form.
2. The asset type MUST be vocabulary, taxonomy, ontology, lexicon or mapping. The word "module" MUST NOT be used.

3. Core assets MUST NOT hold customer-specific or enterprise-specific content. Onboarding a customer MUST add no triples to a core or domain graph.
4. A name MUST NOT contain a customer's name, an environment, a version, a date, a status or a repository name.
5. A published asset name MUST NOT change. A rename is a new asset, and the old one is retired with its replacement recorded, as the Metadata Standard's retirement rule already requires.
6. The current-release graph IRI MUST equal the asset URI. Archived releases MUST load into version IRIs.
7. A tenant ID MUST be opaque, issued once at provisioning, and never reused.
8. Services MUST derive the graphs a caller may reach from the caller's tenant and persona. A service MUST NOT accept a graph IRI from a caller or a screen.
9. A lexicon MUST declare, in its catalog record, the assets whose concepts it labels. Label records MUST point to concepts by IRI and MUST NOT copy them.
10. A domain asset SHOULD declare its parent core assets in its metadata. Parentage is not encoded in the name.
11. Operational domains follow the domain tier. The first word of the name identifies the domain.
12. A draft graph MUST be deleted when its change is released or rejected.
13. A prefix MUST be lowercase and unique across repositories. A customer-tier prefix SHOULD begin with the tenant ID.

Examples

The subjects and the tenant ID are illustrative. The asset URI stands for the Guide's URI pattern ending in the asset name.

Example	What it is
`vocabulary-core-party`	A core vocabulary
`taxonomy-domain-workforce-advisory`	A domain taxonomy
`ontology-domain-payroll-operations`	An operational-domain ontology
`lexicon-domain-workforce-advisory`	Labels common to one domain
`lexicon-enterprise-insperity`	Insperity house terms

Example	What it is
`lexicon-customer-t7f3a9c21`	One tenant's labels
`vocabulary-customer-t7f3a9c21-plan-names`	One tenant's local concepts
`mapping-customer-t7f3a9c21-plan-names-to-benefit-classes`	One tenant's match relations to domain concepts
Asset URI + `/1.2.0`	The graph holding the archived release 1.2.0
Asset URI + `/draft/chg-0412`	The working copy of change chg-0412
Asset URI + `/changes`	The change records of that asset
`skl-core-domain`, `skl-lexicons`, `skl-extensions`	Shared repositories
`skl-tenant-t7f3a9c21`	A dedicated repository for one tenant

Amendments to existing text and open decisions

The insert changes five things in the Metadata Standard as it stands.

1. **Term labels.** The Standard says a term can carry any number of alternate, acronym, persona and hidden labels. In core and domain vocabularies those labels become the vetted universal set only. Labels valid for the enterprise or for one tenant live in lexicons.
2. **Asset types.** Lexicon and mapping join vocabulary, taxonomy and ontology. Each needs its own metadata header, like the existing three.
3. **New fields.** Tier; tenant ID (customer tier only); for a lexicon, the assets it labels; for a mapping, its source and target assets. The properties are chosen when the headers are written.
4. **Catalog record.** It stays the home of the named graph and repository, and also records tier and tenant ID. The tenant-to-organization mapping is restricted.
5. **Glossary.** Add lexicon, mapping, tier, tenant ID and draft graph. The single-letter codes (T, V, X, O) can take L for lexicon and M for mapping.

Open decisions:

- Is `t` plus 8 hex characters the right tenant ID, and what is the name length limit?

- Do persona-specific labels stay on terms when they hold for every tenant, and move to a lexicon when they do not?
- Is an ontology ever allowed at the enterprise tier, for Insperity-wide ontologies?
- Is there one catalog graph, or one per repository?
- What path pattern does the Guide's URI rule give lexicon and mapping assets?

Sources: the Semantic Asset Metadata Standard (15 October 2026), for the graph IRI, version IRI, catalog record, repository, prefix and retirement rules this insert builds on.